



Mid-Term Outcomes and Outcome Predictors after Radiofrequency Ablation for Indeterminate Thyroid Nodules Following Functional Imaging Screening: A Retrospective Cohort Study

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Objective: Current guidelines recommend surveillance or diagnostic lobectomy for indeterminate thyroid nodules. Although radiofrequency ablation (RFA) has emerged as a potential alternative, its role remains controversial. Functional imaging modalities, including 99mTc-methoxyisobutylisonitrile scintigraphy and ¹⁸F-fluorodeoxyglucose positron emission tomography, have been used to support risk stratification. This study evaluated the mid-term outcomes of RFA and their predictors in a functionally screened cohort of patients with indeterminate thyroid nodules.

Materials and Methods: This retrospective cohort study included 109 patients with Bethesda category III or IV thyroid nodules who underwent RFA after functional imaging-based evaluation between January 2018 and November 2024. The primary outcome was local treatment failure (LTF), defined as the first occurrence of nodule regrowth, an increase in vital volume, or repeated indeterminate cytology. The secondary outcome was complete disappearance of the nodule. Supplementary outcomes included the volume reduction rate, symptom and cosmetic scores, and biochemical markers.

Results: During a median follow-up of 2.2 years (range, 0.7–6.8 years), LTF occurred in 15 of 109 patients (13.8%), whereas complete disappearance was observed in 9 (8.3%) patients. The 3-year LTF-free probability and cumulative incidence of complete disappearance were 85.4% and 6.5%, respectively. Significant reductions were observed in the nodule volume, symptom scores, cosmetic scores, and free T4 and thyroglobulin levels. A greater initial maximum nodule diameter was independently associated with a higher likelihood of LTF (hazard ratio [HR] per 1-cm increase, 2.35; 95% confidence interval [CI], 1.23–4.48; $P = 0.010$) and a lower likelihood of complete disappearance (HR, 0.16; 95% CI, 0.04–0.60; $P = 0.006$). At 3 years, 93% of the tumors with an initial diameter <3.0 cm remained free of LTF, whereas no tumors >2.1 cm achieved complete disappearance.

Conclusion: In this functionally screened cohort of patients with indeterminate thyroid nodules, RFA demonstrated acceptable outcomes with volume reduction and symptom relief, particularly in patients with nodules <3 cm.

Keywords: Radiofrequency ablation; Indeterminate thyroid nodules; 99mTc-MIBI scintigraphy; FDG-PET/CT; Outcome; Local treatment failure; Complete regression; Prediction; Predictor

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INTRODUCTION

The management of indeterminate thyroid nodules classified as Bethesda category III or IV remains a clinical challenge. Current guidelines recommend a stepwise risk-adapted approach that includes repeated fine-needle aspiration (FNA), molecular testing when available, and diagnostic lobectomy. Major societies, including the American Thyroid Association and European Thyroid Association, emphasize that image-guided ablation techniques, such as radiofrequency ablation (RFA), should generally be limited to nodules with confirmed benign cytology. Accordingly, surgery remains the recommended first-line treatment for indeterminate nodules owing to the residual risk of malignancy and lack of robust prospective outcome data [1-6].

Nevertheless, RFA has gained increasing attention as a minimally invasive alternative for carefully selected patients who are unsuitable for or refuse surgery. Retrospective studies and meta-analyses have reported substantial volume reductions, low complication rates, and favorable clinical outcomes when stringent selection criteria were applied [7-9]. These findings suggest that RFA may be considered a feasible treatment option for select indeterminate nodules with a low likelihood of malignancy.

Functional imaging modalities, including ^{18}F -fluorodeoxyglucose positron emission tomography (FDG-PET) and technetium-99m methoxyisobutylisonitrile (99mTc-MIBI) scintigraphy, have been investigated as adjunct tools for risk stratification. Randomized trials have demonstrated that FDG-PET-guided management can reduce unnecessary diagnostic surgery for benign nodules by approximately 40%, whereas a negative FDG-PET result demonstrates a high negative predictive value (NPV) that may support surveillance in appropriately selected patients [10-12]. Similarly, quantitative 99mTc-MIBI imaging using washout index analysis demonstrated a promising diagnostic performance in a small prospective cohort. However, these techniques remain incompletely validated and lack standardized thresholds, limiting their widespread adoption [13-15].

Despite these advances, evidence regarding RFA outcomes in a functionally screened population remains limited. To date, only one small study has reported the outcomes of RFA in 22 Bethesda IV nodules with negative FDG-PET findings, demonstrating favorable short-term results [16]. However, data on the mid-term durability and predictors of treatment success remain limited.

Therefore, this study aimed to evaluate the mid-term outcomes of RFA in indeterminate thyroid nodules following functional imaging-based evaluation, and identify predictors of outcomes that may help inform patient selection.

MATERIALS AND METHODS

This study was approved by the Institutional Review Board of Kaohsiung Chang Gung Memorial Hospital (IRB No. 202501493B0).

Study Protocol

This retrospective cohort study included patients who underwent RFA for thyroid nodules with repeated indeterminate cytology (Bethesda category III or IV) at the Kaohsiung Chang Gung Memorial Hospital between 2018 and 2024. Repeat FNA was routinely performed for nodules initially classified as Bethesda category III or IV. All included patients demonstrated indeterminate cytology on at least two separate cytological assessments, including the one performed at our institution. Inclusion criteria required pre-RFA functional imaging demonstrating either a 99mTc-MIBI washout index ≤ 40 or an ^{18}F -FDG PET/CT maximum standardized uptake value (SUVmax) < 5 . Patients were excluded if they lacked pre-RFA functional imaging, had a follow-up duration < 0.5 years, or presented with an initial nodule diameter < 1.5 cm. In routine clinical practice, all patients with Bethesda category III or IV cytology are counseled regarding management options, including active surveillance, surgery, and RFA. Therefore, the study cohort represented a functionally screened subgroup that met the predefined imaging criteria and elected to undergo RFA after shared decision-making, typically declining surgical intervention.

Among 222 patients with Bethesda category III or IV

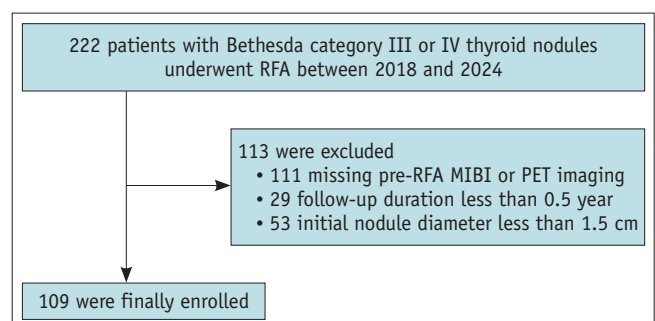


Fig. 1. Flowchart of patient enrollment and selection for RFA. RFA = radiofrequency ablation, MIBI = methoxyisobutylisonitrile

thyroid nodules who underwent RFA during the study period, 113 were excluded owing to missing pre-RFA MIBI or PET imaging ($n = 111$), follow-up <0.5 years ($n = 29$), and initial nodule diameter <1.5 cm ($n = 53$) (Fig. 1). The final cohort comprised 109 patients (mean age, 45.1 ± 12.4 years; 95 female).

Patients were followed up at 3, 6, and 12 months after RFA, and annually thereafter. Clinical evaluations included symptom and cosmetic scores and ultrasound-based nodule volume measurement using the ellipsoid formula (length $\times$ width $\times$ height $\times \pi/6$). The symptom score ranged from 0 to 5, with one point assigned for each symptom (compression, cough, dysphagia, voice change, and pain). The cosmetic score ranged from 0 to 3 (0 = no palpable mass; 1 = palpable mass without cosmetic concern; 2 = cosmetic concern during neck extension or swallowing; and 3 = readily visible cosmetic problem) [17]. Laboratory assessments were performed at all follow-up visits except for the 3-month visit and included total triiodothyronine (T3), free thyroxine (free T4), thyroid-stimulating hormone (TSH), thyroglobulin (TG), anti-thyroid peroxidase antibody (anti-TPO), TG antibody (anti-Tg), and anti-TSH receptor antibody (anti-TSHR). Follow-up FNA was recommended 6–12 months after RFA to reassess the cytological status.

Functional Imaging Assessment

All patients underwent pre-RFA functional imaging with either ^{99m}Tc -MIBI scintigraphy or ^{18}F -FDG PET/CT as part of the institutional risk stratification to reduce the likelihood of treating nodules with a higher malignant potential. Nodules <1.5 cm were excluded to maintain quantitative imaging reliability because measurements in small lesions are susceptible to partial volume effects.

For ^{99m}Tc -MIBI scintigraphy, early and delayed images were obtained to calculate the washout index, reflecting tracer retention. Malignant nodules tend to demonstrate persistent uptake, whereas benign nodules typically exhibit substantial washout [13,18,19]. Therefore, a washout index threshold of ≤ 40 was selected as a conservative criterion to maximize specificity for benign disease. For ^{18}F -FDG PET/CT, metabolic activity was assessed using the SUVmax. Previous studies have demonstrated that lower SUVmax values are associated with a lower likelihood of malignancy, with thresholds near 4–5 commonly used to exclude higher-risk nodules [20–24]. Accordingly, an SUVmax cutoff value of <5 was applied.

RFA Devices and Technique

All RFA procedures were performed under local anesthesia using hydrodissection with 5% dextrose to protect adjacent critical structures. A freehand trans-isthmic approach using the moving shot technique was applied under continuous ultrasound guidance. An internally cooled 18-gauge electrode (7-cm shaft; 5-, 7-, or 10-mm active tip) was selected according to the nodule size and perithyroidal anatomy. Ablation was performed using either the VIVA system (STARmed; <https://starmed-america.com/viva-combo-rf-system>) or M2004 generator (RF Medical; https://rfmedical.en.ec21.com/RF_Generator-M2004--1231492_4941109.html).

Outcome Assessments

The primary outcome was local treatment failure (LTF), defined as an incomplete local response after RFA. LTF was identified based on the first occurrence of any of the following components: 1) nodule regrowth, 2) increase in vital volume (Vv), or 3) repeated indeterminate cytology during follow-up FNA. However, these components are not mutually exclusive. The total volume (Vt) refers to the entire nodule volume, ablated volume (Va) represents the treated region, and Vv is calculated as Vt minus Va. Nodule regrowth was defined as a $\geq 50\%$ increase in Vt relative to the smallest previously recorded volume, and Vv increase was defined analogously. Repeated indeterminate cytology was defined as continued Bethesda category III or IV classification after RFA. Ultrasound-based components were assessed in the entire cohort, whereas cytology-based non-resolution components were evaluated only in patients who underwent post-RFA FNA.

The secondary outcome was the complete disappearance of the treated nodule on imaging during follow-up. The supplementary outcomes included longitudinal changes in nodule volume, volume reduction rate (VRR), symptoms, cosmetic scores, and thyroid-related laboratory parameters.

Statistical Analysis

All statistical analyses were conducted using R software version 4.0.3 (<https://www.r-project.org>).

The baseline characteristics were summarized for the overall cohort. Categorical variables were reported as counts and percentages, whereas continuous variables are presented as means $\pm$ standard deviations or medians with interquartile ranges, as appropriate. Longitudinal outcomes were analyzed using the Skillings–Mack test with Bonferroni-adjusted post-hoc comparisons. Statistical

Table 1. Baseline characteristics of patients with indeterminate thyroid nodules treated with RFA

Characteristic	Overall (n = 109)
Age, yr	
Mean ± SD	45.1 ± 12.4
Sex	
Female	95 (87.2)
Male	14 (12.8)
Maximum diameter, cm	
Mean ± SD	3.20 ± 1.29
Median [Q1, Q3]	3.00 [2.10, 4.20]
Lesion site	
Right lobe	56 (51.4)
Left lobe	46 (42.2)
Isthmus	7 (6.4)
Bethesda categories	
Bethesda III	40 (36.7)
Bethesda IV	69 (63.3)
MIBI washout index	
Median [Q1, Q3]	-49.20 [-55.55, -46.00]
Missing	50 (45.9)
PET maximum SUV	
Median [Q1, Q3]	2.40 [1.88, 3.01]
Missing	62 (56.9)
Total energy in RFA, Kcal	
Median [Q1, Q3]	3.79 [2.65, 6.62]
Missing	17 (15.6)
Symptom score	
Median [Q1, Q3]	0 [0, 2.00]
Cosmetic score	
Median [Q1, Q3]	3.00 [2.00, 3.00]
Free T4, ng/dL	
Mean ± SD	1.10 ± 0.56
Missing	6 (5.5)
T3, ng/dL	
Mean ± SD	91.90 ± 14.51
Missing	9 (8.3)
TSH, uIU/mL	
Median [Q1, Q3]	1.07 [0.73, 1.54]
Missing	5 (4.6)
Anti-TSHR, IU/L	
Median [Q1, Q3]	0.80 [0.80, 0.90]
Missing	37 (33.9)
Anti-TPO, IU/mL	
Median [Q1, Q3]	3.00 [1.07, 3.56]
Missing	36 (33.0)
Anti-THYG, IU/mL	
Median [Q1, Q3]	18.10 [14.62, 25.90]
Missing	95 (87.2)

Table 1. Baseline characteristics of patients with indeterminate thyroid nodules treated with RFA (continued)

Characteristic	Overall (n = 109)
TG, ng/mL	
Median [Q1, Q3]	60.90 [15.35, 188.75]
Missing	37 (33.9)

Data are presented as number of patients with percentage in parentheses, unless specified otherwise. "Missing" indicates the number of patients for whom the corresponding data were not available.

RFA = radiofrequency ablation, SD = standard deviation, Q1 = 25th percentile, Q3 = 75th percentile, Bethesda III = atypia of undetermined significance/follicular lesion of undetermined significance, Bethesda IV = follicular neoplasm or suspicious for a follicular neoplasm, MIBI = methoxyisobutylisonitrile, SUV = standardized uptake value, Free T4 = free thyroxine, T3 = triiodothyronine, TSH = thyroid-stimulating hormone, TSHR = thyroid-stimulating hormone receptor antibody, TPO = thyroid peroxidase antibody, THYG = thyroglobulin antibody, TG = thyroglobulin

significance was defined as a two-sided *P*-value of <0.05.

Time-to-event outcomes were analyzed using Kaplan–Meier estimation and Cox proportional hazards regression, accounting for variable follow-up durations and right censoring. Predictors of LTF and complete disappearance of the nodule were evaluated using multivariable Cox regression analysis with hazard ratios (HRs) and 95% confidence intervals (CIs). Time-dependent receiver operating characteristic (ROC) analysis was performed 3 years after RFA. The optimal threshold values for predicting LTF and complete tumor disappearance were chosen as the values maximizing the F1 score.

RESULTS

Patient Characteristics

A total of 109 patients were included in the final cohort, with a median follow-up duration of 2.2 years (range, 0.7–6.8). The median initial maximum nodule diameter was 3.0 cm (range, 1.5–8.8 cm). The baseline characteristics of the study cohort are summarized in Table 1.

Primary and Secondary Outcomes

LTF occurred in 15 patients (13.8%), including 10 with repeated indeterminate cytology on follow-up FNA, 5 with nodule regrowth, and 1 with an increase in Vv; one patient met both cytologic and volumetric criteria. Complete disappearance of the nodule was observed in 9 patients (8.3%). Kaplan–Meier analysis demonstrated a 3-year LTF-free probability of 85.4% (95% CI, 78.6%–92.8%) (Fig. 2A).

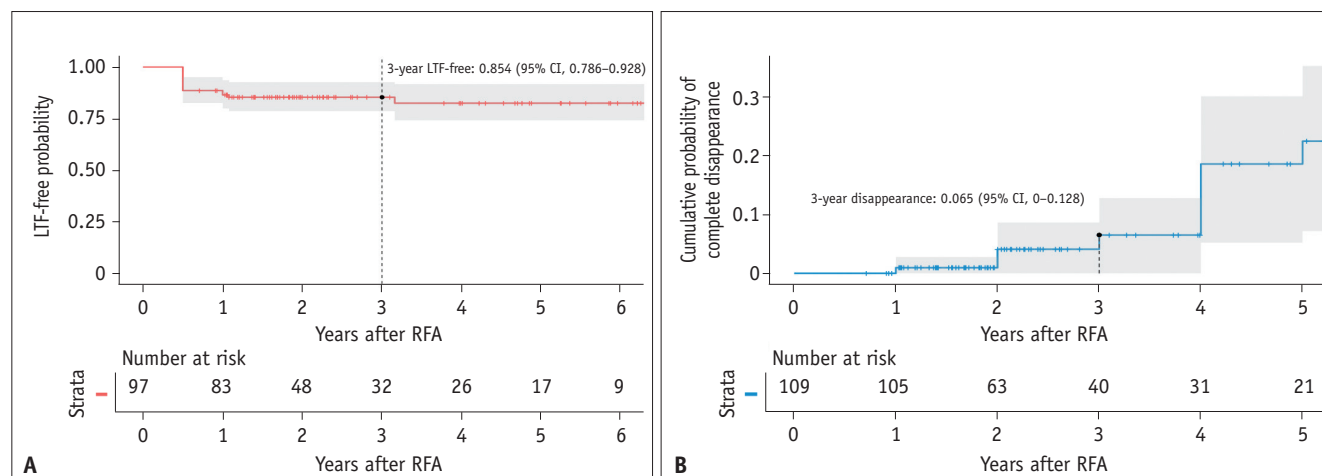


Fig. 2. Kaplan–Meier curves showing time-to-event outcomes after RFA for indeterminate thyroid nodules. **A:** LTF-free probability over time. **B:** Cumulative incidence of complete disappearance of the nodule over time. Dashed vertical lines indicate the 3-year time point, and solid dots denote the corresponding Kaplan–Meier estimates at 3 years with 95% CIs. Shaded areas represent 95% CIs. The number of patients at risk at each time point is shown below the plots. RFA = radiofrequency ablation, LTF = local treatment failure, CI = confidence interval

The 3-year cumulative incidence of complete disappearance of the nodule was 6.5% (95% CI, 0%–12.8%) (Fig. 2B).

Among the five patients with nodule regrowth, three underwent repeat RFA and two were managed with active surveillance. None of the patients required surgical intervention, and all the nodules subsequently demonstrated stabilization or further volume reduction during follow-up. No local or distant metastases were observed during the follow-up period. Post-RFA follow-up FNA was performed in 97 of 109 patients. The cytological results were Bethesda category I in 64 patients, category II in 23, category III in 8, and category IV in 2. None of the post-RFA FNA results were classified as Bethesda category V or VI. Regarding treatment exposure, 86 patients underwent a single RFA session, 21 underwent two sessions, and 2 underwent three sessions.

Safety and Complications

Procedure-related complications were rare. Minor complications occurred in 4 of the 109 patients (3.7%), and 1 major complication (0.9%) was observed. All complications included voice hoarseness associated with vocal cord palsy confirmed on laryngoscopic examination. Among the five affected patients, four demonstrated complete recovery of vocal cord mobility on follow-up laryngoscopy at 1–3 months (1, 1, 2, and 3 months). One patient exhibited persistent vocal cord palsy on follow-up laryngoscopy, with partial improvement at 3 months and was therefore classified as a major complication despite

mild symptoms.

Longitudinal Changes in the Clinical and Laboratory Outcomes

Significant longitudinal reductions were observed in the nodule volume, symptom scores, cosmetic scores, and free T4 and TG levels (all $P < 0.05$) (Fig. 3). The nodule volume significantly declined across consecutive follow-up intervals of 36 months, with a VRR of 56% at 3 months and 91% at 3 years. Symptoms and cosmetic scores exhibited parallel improvement over time. Thyroid function parameters, including T3 and TSH levels and anti-thyroid antibodies (anti-TPO, anti-Tg, and anti-TSHR) remained stable throughout the follow-up period.

Predictors of LTF and Complete Disappearance of the Nodule

In multivariable Cox proportional hazards analysis (Table 2), initial maximum nodule diameter was an independent predictor of LTF after adjustment for age, sex, initial Bethesda category, and MIBI index (HR per 1-cm increase, 2.35; 95% CI, 1.23–4.48; $P = 0.010$). Bethesda category IV exhibited a tendency of a higher rate of LTF compared with category III with a marginal P -value, not reaching statistical significance (HR, 3.44; 95% CI, 0.87–13.56; $P = 0.077$). For complete disappearance of the nodule, a greater initial maximum diameter was independently associated with a lower likelihood of the outcome (HR, 0.16; 95% CI, 0.04–0.60; $P = 0.006$). As the event represents a favorable outcome, HR <1 reflects

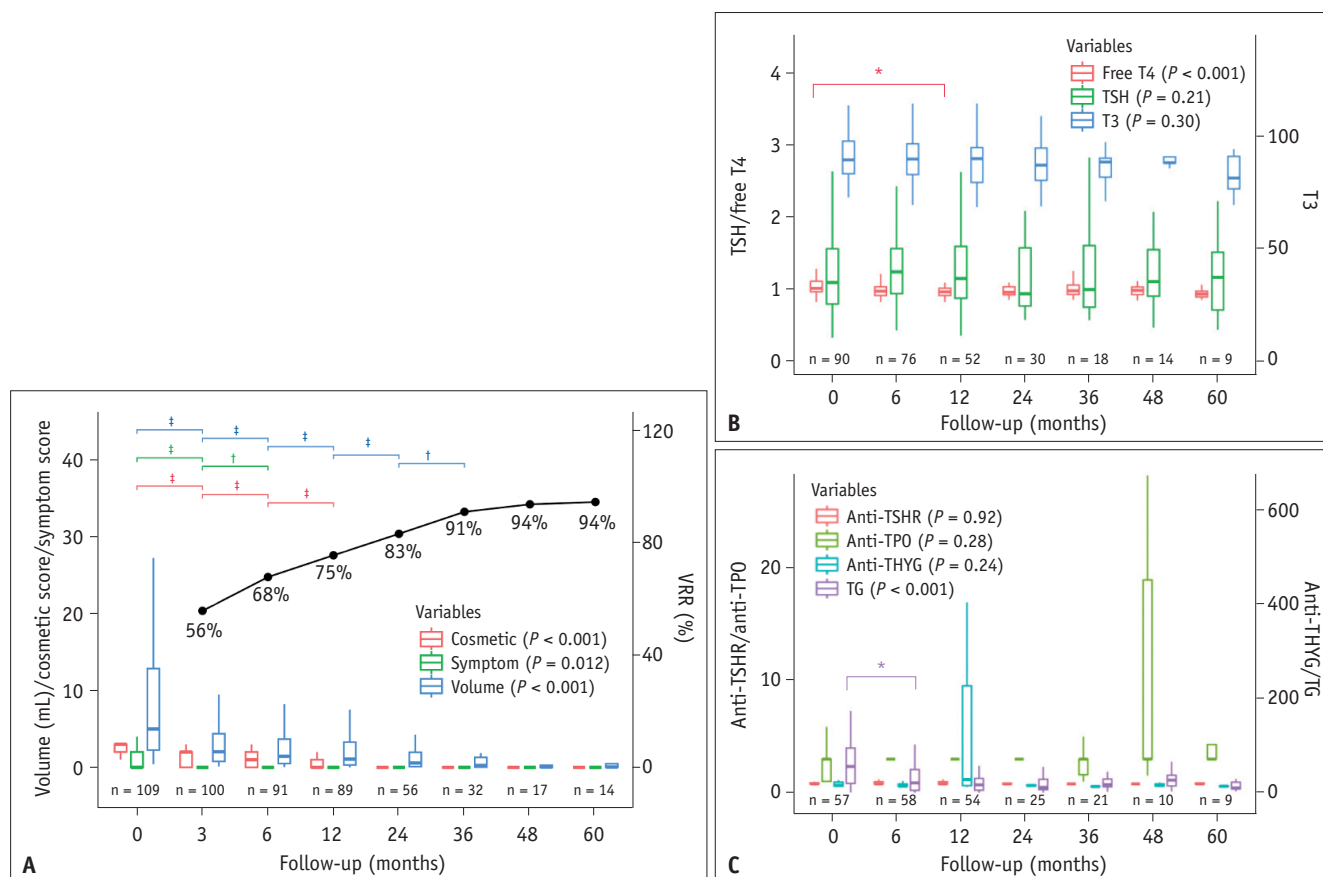


Fig. 3. Longitudinal changes in the clinical outcomes and laboratory parameters from baseline to 60 months after radiofrequency ablation. **A:** Changes in the nodule volume, cosmetic score, symptom score, and VRR. **B:** Changes in the thyroid function tests, including serum TSH, free T4, and T3 levels. **C:** Changes in anti-thyroid antibodies (anti-TSH receptor antibody, anti-TPO, anti-TG antibody) and TG levels. Asterisks denote statistically significant differences between adjacent time points (* $P < 0.05$, $^{\dagger}P < 0.01$, $^{\ddagger}P < 0.0001$). Sample size at each time point is shown below the x-axis. VRR = volume reduction rate, TSH = thyroid-stimulating hormone, TPO = thyroid peroxidase antibody, TG = thyroglobulin, TSHR = thyroid-stimulating hormone receptor antibody

a decreased probability of complete disappearance with increasing size.

Time-dependent ROC curve analysis at 3 years demonstrated that the initial maximum nodule diameter exhibited a modest discriminative ability for predicting LTF (area under the curve [AUC], 0.63; 95% CI, 0.45–0.82). A criterion of ≥ 3.0 cm for LTF prediction yielded a sensitivity of 0.72, specificity of 0.65, positive predictive value (PPV) of 0.26, and NPV of 0.93 (i.e., 93% of tumors with an initial diameter < 3.0 cm will not develop LTF). The discriminative performance was higher for complete disappearance (AUC, 0.80; 95% CI, 0.66–0.95). A criterion of ≤ 2.1 cm for predicting complete tumor disappearance achieved a sensitivity of 1, specificity of 0.74, PPV of 0.14, and NPV of 1 (i.e., initial diameter > 2.1 cm will not show complete disappearance).

Figure 4 presents a representative case demonstrating

durable complete disappearance after RFA, and Figure 5 illustrates a case of early regrowth followed by subsequent volume reduction.

DISCUSSION

To the best of our knowledge, this study represents one of the largest cohort studies evaluating the mid-term time-to-event outcomes of RFA for indeterminate thyroid nodules within a functionally screened population. Among 109 patients with Bethesda category III or IV nodules ≥ 1.5 cm, all of whom underwent either 99mTc-MIBI or FDG-PET screening prior to ablation, the 3-years LTF-free probability was 85.4%, and the cumulative incidence of complete disappearance at 3 years was 6.5%. Significant improvements were observed in symptoms and cosmetic scores, TG and T4 levels, and nodule volume, with a VRR

Table 2. Multivariable Cox proportional hazards regression models for local treatment failures and complete nodule disappearance

Variable	HRs (95% CI)	P
Local treatment failure		
Age	1.02 (0.97–1.07)	0.446
Sex, male vs. female*	0.38 (0.04–3.43)	0.392
Maximum diameter	2.35 (1.23–4.48)	0.010
Bethesda IV vs. III*	3.44 (0.87–13.56)	0.077
MIBI washout index	0.98 (0.92–1.03)	0.395
Complete nodule disappearance		
Age	1.01 (0.94–1.09)	0.741
Sex, male vs. female*	0.16 (0.01–1.78)	0.136
Maximum diameter	0.16 (0.04–0.60)	0.006
Bethesda IV vs. III*	0.31 (0.02–4.10)	0.373

HRs for continuous variables are reported per unit increase (age per 1 year; maximum diameter per 1 cm; MIBI washout index per 1 unit). HRs >1 and <1 indicate higher and lower occurrence rates of the events compared to the base state, respectively.

*Base category.

HR = hazard ratio, MIBI = methoxyisobutylisonitrile, CI = confidence interval, Bethesda IV = follicular neoplasm or suspicious for a follicular neoplasm, Bethesda III = atypia of undetermined significance/follicular lesion of undetermined significance

of 91% at 3 years. Importantly, baseline maximum nodule diameter emerged as an independent predictor of both LTF and complete disappearance, with thresholds of ≥ 3.0 cm for predicting LTF and ≤ 2.1 cm for predicting complete tumor disappearance, both yielding high NPV of 93% and 100%, respectively.

The potential clinical role of thermal ablation for indeterminate thyroid nodules warrants further consideration. Unlike benign nodules, where the primary goal of RFA is symptomatic relief, the therapeutic objective for indeterminate nodules may include achieving sufficient ablation to control lesions with uncertain malignant potential. Therefore, RFA may serve as a minimally invasive alternative to diagnostic thyroid surgery in carefully selected patients. By providing local tumor control while avoiding surgical morbidity, this approach may help reduce unnecessary thyroid surgery for nodules that ultimately behave in a benign manner.

Our findings align with those of previous studies demonstrating sustained volume reduction after RFA for indeterminate thyroid nodules by demonstrating that similar outcomes are observed when RFA is integrated with functional imaging screening. A meta-analysis by Lim et al. [7] reported a pooled 12-month VRR of 81% with stabilization thereafter, which was comparable to the 75% VRR at 12 months observed in our cohort. Previous studies

have also reported higher disappearance rates in smaller nodules, supporting baseline size as an important prognostic factor [25,26]. Consistent with these observations, smaller initial maximum diameter independently predicted complete disappearance of the nodule in our cohort (HR, 0.16; 95% CI, 0.04–0.60; $P = 0.006$). Taken together, these findings suggest that the effectiveness of RFA in our study is broadly comparable to that reported in prior literature, even within a functionally screened population.

With respect to safety, prior reports have described progression or regrowth rates ranging from approximately 1.9% to 15.2% over 0.5 to 3 years of follow-up, with low rates of delayed surgery and major complications [26–34]. The LTF rate in the present study was within this range. These differences likely reflect heterogeneity in the study design and follow-up duration. Notably, we adopted a composite endpoint that incorporated regrowth, Vv increase, and repeated indeterminate cytology to capture clinically meaningful incomplete treatment responses. As indeterminate cytology may perpetuate diagnostic uncertainty and patient anxiety, this definition was intended to reflect scenarios in which the therapeutic objective of resolving uncertainty was not fully achieved. In addition, a diameter criterion of ≥ 3.0 cm yielded a high NPV (93%) for LTF, suggesting a low likelihood of incomplete response among the smaller nodules within this functionally screened cohort. These findings are consistent with prior evidence demonstrating minimal regrowth or progression in nodules <2–3 cm [25,26], and support the use of baseline size as a practical factor when counseling patients regarding expected outcomes. Another important observation in our cohort was the absence of malignant cytology or metastatic disease during the follow-up. Among the 97 patients who underwent post-RFA FNA, none were classified as Bethesda category V or VI cytology, and no local or distant metastases were detected. These findings suggest that functional imaging-based selection may help identify nodules with a low probability of clinically significant malignancy.

Lim et al. [7] highlighted in their meta-analysis that incorporating advanced diagnostic tools, such as molecular analysis and radioisotope imaging, into pretreatment evaluations may help identify nodules with a lower likelihood of malignancy. Building on this concept, our findings suggest a practical framework incorporating functional imaging and baseline nodule size. In our cohort, a diameter threshold of 3.0 cm demonstrated meaningful risk stratification: nodules ≥ 3.0 cm showed a higher risk of

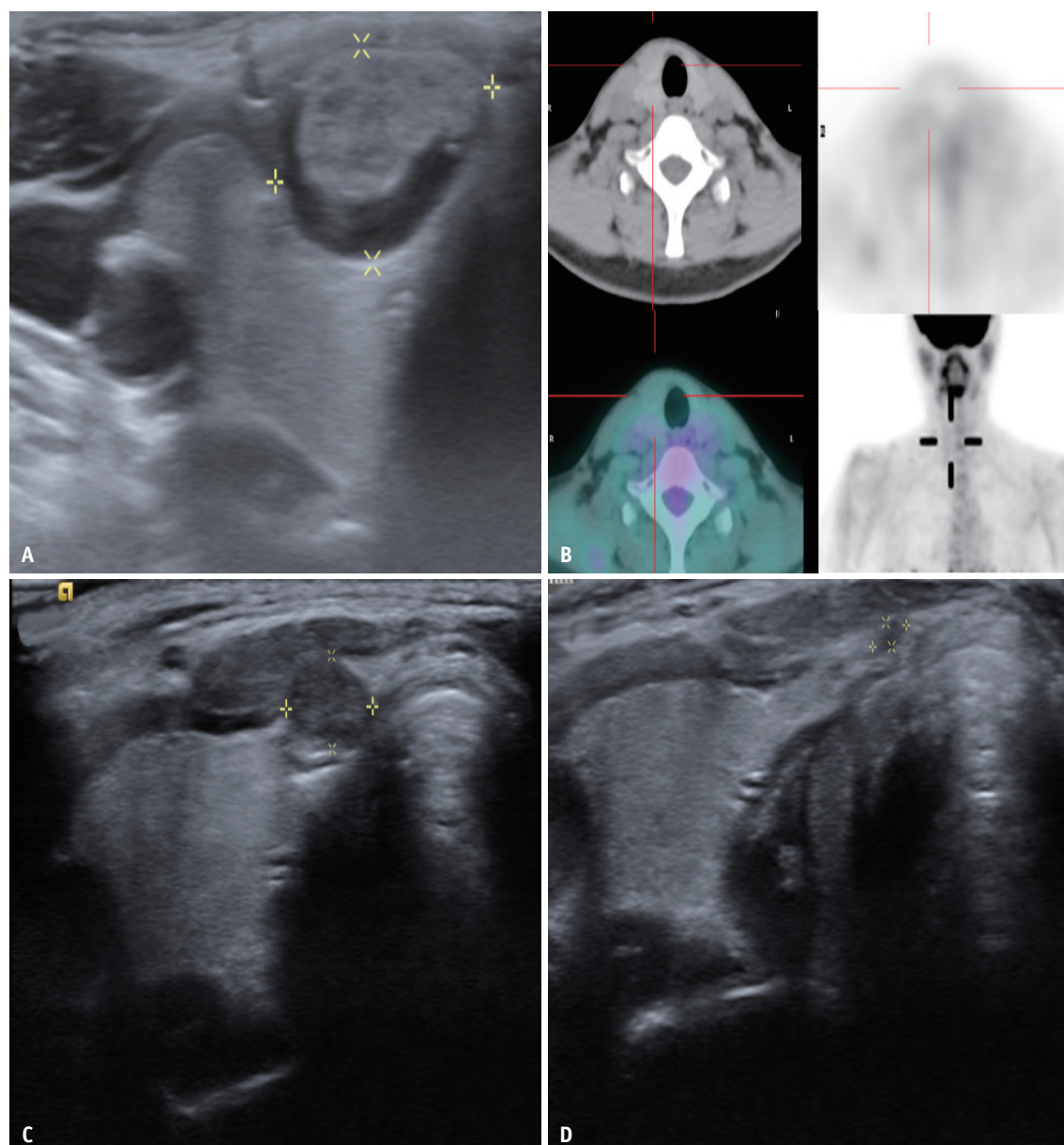


Fig. 4. A 41-year-old woman with an indeterminate thyroid nodule (initial Bethesda category IV), FDG-PET standardized uptake value of 1.4, and a maximum nodule diameter of 1.9 cm. The nodule demonstrated near-complete disappearance at 4 years after RFA, which was sustained during 6.6 years of follow-up. **A-D:** Pre-RFA ultrasound image (**A**) pre-RFA FDG-PET showing low metabolic activity (**B**), post-RFA ultrasound at 2 years (**C**), post-RFA ultrasound at 4 years (**D**). FDG = fluorodeoxyglucose, RFA = radiofrequency ablation

LTF, whereas smaller nodules exhibited a lower likelihood of incomplete response. Notably, nodules measuring ≤ 2.1 cm were more likely to achieve complete disappearance. These findings suggest that the baseline nodule size may help inform patient selection and counseling when considering RFA within a functionally screened population.

This study had certain limitations. First, this retrospective study included a highly selected population. Many patients were excluded based on predefined functional imaging

criteria, nodule size thresholds, and follow-up duration. Only individuals who declined surgery after shared decision-making underwent RFA. Consequently, a selection bias is likely, and the findings may not be generalizable to all patients with indeterminate thyroid nodules. Second, complications, delayed surgery, and final pathology were not completely captured, and complete ablation was not systematically evaluated. Advanced techniques, such as contrast-enhanced ultrasound, microvascular flow imaging,

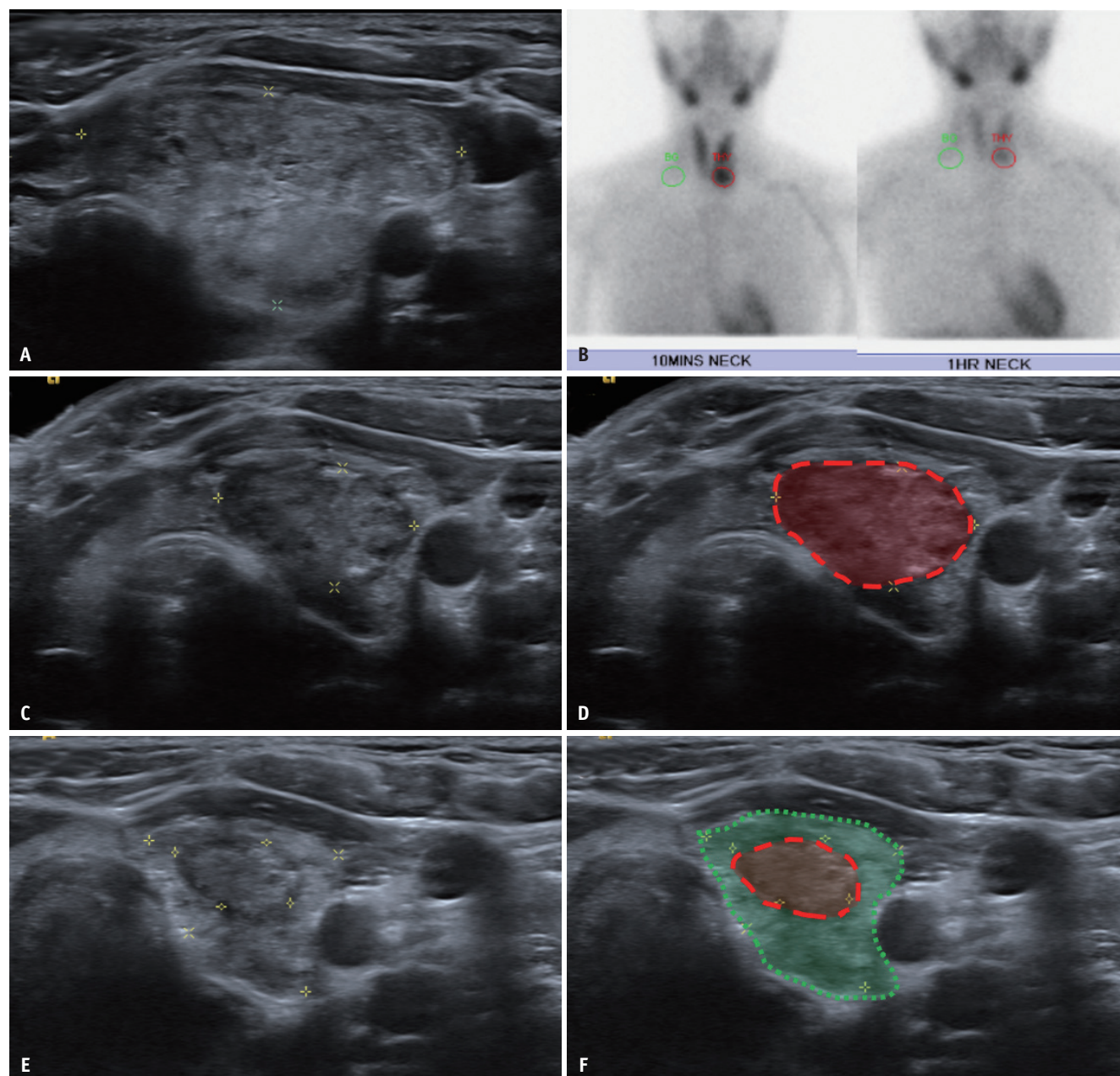


Fig. 5. A 30-years-old woman with an indeterminate thyroid nodule (initial Bethesda category III), a 99mTc-MIBI washout index of -68.7, and a maximum nodule diameter of 3.8 cm. The nodule showed regrowth at 6 months after RFA, followed by subsequent volume reduction during continued follow-up. **A-F:** Pre-RFA ultrasound image (**A**), pre-RFA 99mTc-MIBI scintigraphy (**B**), post-RFA ultrasound at 3 months demonstrating initial volume reduction (**C**), post-RFA ultrasound at 3 months: red area bounded by dash line demonstrating ablated volume (**D**), post-RFA ultrasound at 6 months demonstrating regrowth (**E**), post-RFA ultrasound at 6 months: green area by dotted line demonstrating total volume and red area bounded by dash line demonstrating ablated volume (**F**). 99mTc-MIBI = technetium-99m methoxyisobutylisonitrile, RFA = radiofrequency ablation

and targeted biopsy, are not routinely used to detect residual viable tissue. Third, outcomes were not directly compared between patients with and without functional imaging; therefore, the clinical value of imaging-based screening remains uncertain. Fourth, the proposed diameter thresholds should be considered as exploratory and require

external validation. Fifth, the use of two imaging modalities (FDG-PET and MIBI) introduced potential heterogeneity. Finally, the follow-up period represents mid-term outcomes; a longer follow-up period is required to determine the long-term efficacy and safety of RFA for indeterminate nodules.

In conclusion, this study supports the feasibility of RFA

for the careful selection of indeterminate thyroid nodules. Baseline nodule diameter may serve as a practical factor for patient selection and counseling. Notably, 93% of the nodules smaller than 3.0 cm remained free of LTF, indicating a low likelihood of an incomplete response within this functionally screened cohort. RFA is associated with an acceptable safety profile, sustained volume reduction, and persistent symptom improvement in these patients.

Availability of Data and Material

The datasets generated or analyzed during the study are available from the corresponding author on reasonable request.

Conflicts of Interest

The authors have no potential conflicts of interest to disclose.

Author Contributions

Conceptualization: Wei-Che Lin, Yen-Hsiang Chang. Data curation: Ling-Chieh Shih, An-Ni Lin. Formal analysis: Ling-Chieh Shih. Investigation: Ling-Chieh Shih. Methodology: Ling-Chieh Shih. Project administration: Wei-Che Lin. Resources: Wei-Che Lin, Yen-Hsiang Chang. Software: Ling-Chieh Shih. Supervision: Wei-Che Lin, An-Ni Lin. Visualization: Ling-Chieh Shih. Writing—original draft: Ling-Chieh Shih. Writing—review & editing: Wei-Che Lin, An-Ni Lin.

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None

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